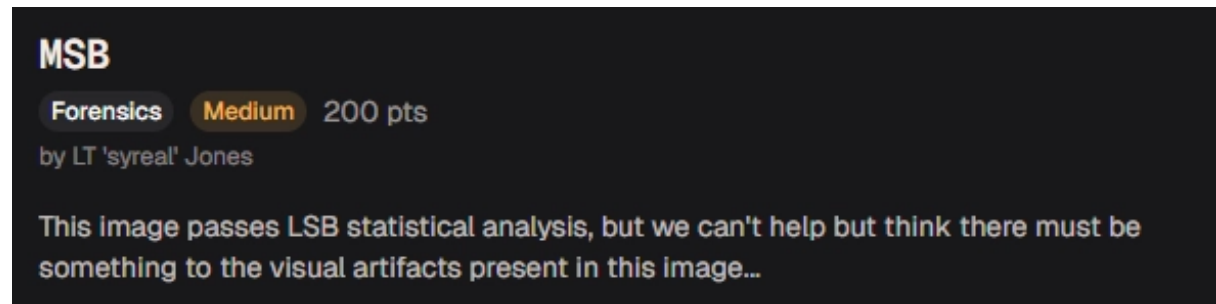


MSB | picoCTF Writeup

written by Alexander Sapo



process:

The challenge hinted at **MSB**, so we performed a quick search and found that it stands for **Most Significant Bit**. In steganography, hidden data is commonly embedded in the **LSB (Least Significant Bit)** because modifying those bits causes minimal visual changes to an image. However, data can also be hidden in the **MSB**, which represents the highest-value bit in a byte. While changing MSBs has a greater impact on pixel values, they can still be used to conceal information.



Since the challenge specifically referenced MSB, we decided to inspect the image's most significant bits instead of the least significant bits.

To do this, we used **StegSolve**:

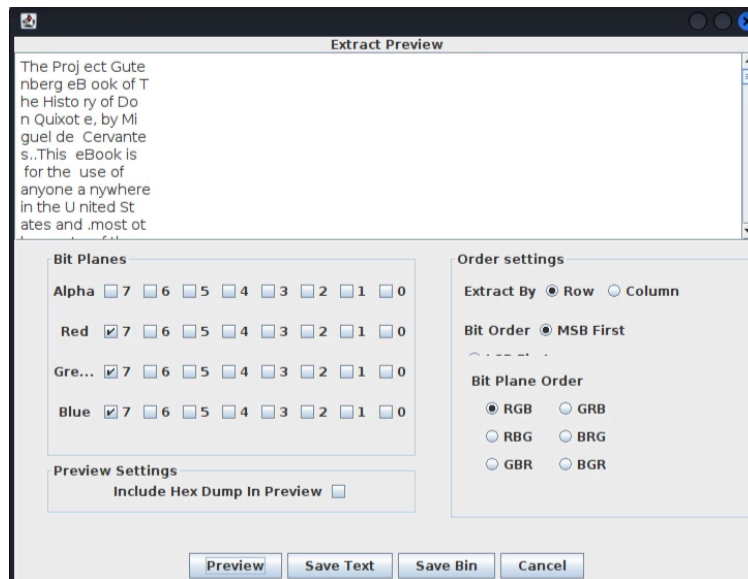
```
java -jar stegsolve.jar
```

After launching the GUI, we navigated to:

Analyse → Data Extract

In the extraction settings, we selected:

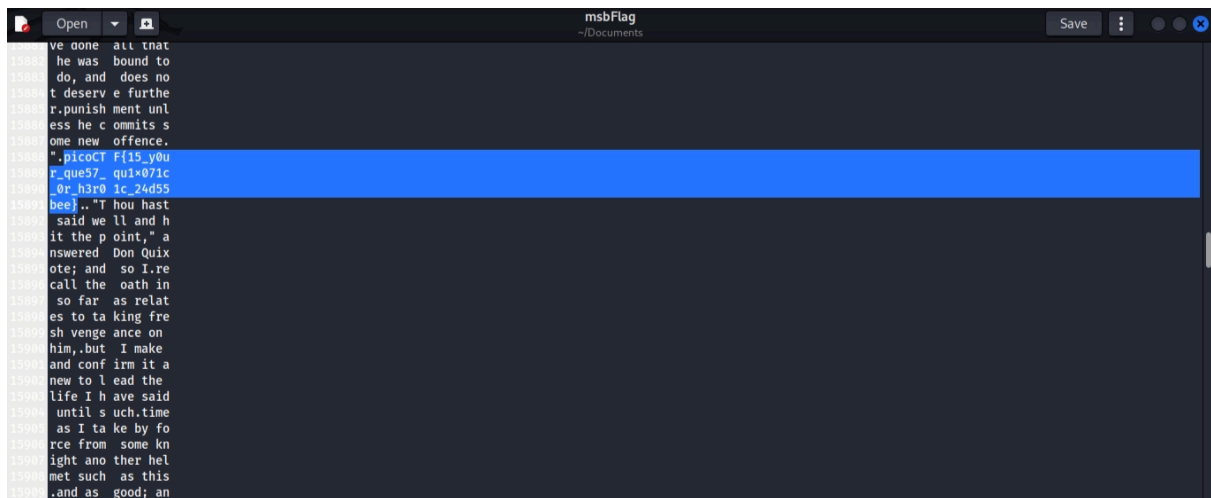
- **Extract by Row**
- **MSB First**
- **Bit Plane Order: RGB**
- Checked the first row of **R**, **G**, and **B**



After clicking **Preview**, a large amount of readable text became visible, indicating that hidden data had been successfully extracted.

We then saved the output and searched for the flag using:

```
strings extracted.txt | grep "pico"
```



This revealed the flag and completed the challenge.

flag: picoCTF{15_y0ur_que57_qu1x071c_0r_h3r01c_24d55bee}